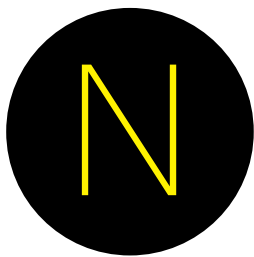




BY PAWAN GOENKA

Even the Sky is Not the Limit Anymore

Public-private partnership in the space sector is helping the Indian space economy grow by leaps and bounds



NOVEMBER 18, 2022, will forever be etched as a memorable day in the history of the Indian space sector thanks to the successful launch of Vikram-S—the first privately built rocket—by Sky-root Aerospace from the Indian Space Research Organisation's spaceport at Satish Dhawan Space Centre in Andhra Pradesh. The success of Mission Prarambh has ushered in a new era for the Indian space sector. It signifies a new dawn for public-private co-operation in enriching the sector.

Going back 50 years, it was the launch of India's first indigenous rocket, Rohini-75, on November 20, 1967, that marked a new start. There has been no looking back since, thanks to the leadership of ISRO and the unwavering support of the government. Today, India is among the top few space-faring nations in the world.

However, we all know that despite being a major space-faring nation, India accounts for only about 2 per cent of the global space economy. This is about to change. With the liberalisation of the sector, new entrants from the private sector have already begun to gain global attention. The surge in space activities with the participation of private players is likely to push up India's ranking. The world is also looking at India as the next big destination in the sector.

Globally, there are going to be several opportunities across domains in the next 25 years. We are entering into a new era for the industry—from extra-terrestrial exploration to declining launch costs—the space sector is truly taking off for newer orbits. In this era, the opportunity for private industry is immense. Per a report from Morgan Stanley, the global space economy is expected to grow to \$1 trillion by 2040.

The Indian space sector, too, is undergoing a sea change post the announcement of the space sector reforms. The unprecedented rise in India's space start-ups hints at a revolution brewing on the horizon. The number of space start-ups in the country has gone up by more than 200 per cent in the past two

years, and they number over 150 now. Traditional MSME vendors to ISRO are also diversifying by going up the value chain. For instance, a private consortium has taken a giant step to manufacture ISRO's workhorse, the polar satellite launch vehicle (PSLV). Investor interest is also growing, with total investments into the sector crossing \$110 million in 2022, which is more than the total investments till 2021. Admittedly, these figures compared to the global numbers are very small, but they reflect the rising confidence of investors in India.

Space activity across segments is gaining momentum in India. We must seize this opportunity to earn our rightful share in the global space economy. While upstream activities are getting higher attention, the midstream and downstream activities are also expanding. We have the potential to lead in downstream applications backed by India's strength in advanced IT and data skills. This expertise can have wide applications in monitoring changing climate, extreme weather, use of natural resources, etc., thereby creating new economic opportunities and positively impacting the lives of people. And thus reducing the space between spacetechnology and people.